

HACKTOBERFEST 2026 • CHALLENGE 1: BUILD FOR A FRIEND

NEET Airplane Tutor (Offline Socratic RAG)

Target Recipient: Sanya (Sister, NEET Medical Aspirant)
Core AI: Local Ollama (llama3.2 + nomic-embed-text)

Submission Deadline: Oct 5, 2026 @ 12:29 PM IST (06:59 UTC)
Stack: Next.js 15, TypeScript, Tailwind, Express (Render)

1. EXECUTIVE SUMMARY & PROBLEM DEFINITION

The Recipient: A high-school student rigorously preparing for the Indian National Eligibility cum Entrance Test (NEET - Physics, Chemistry, Biology).

The Core Friction Points:

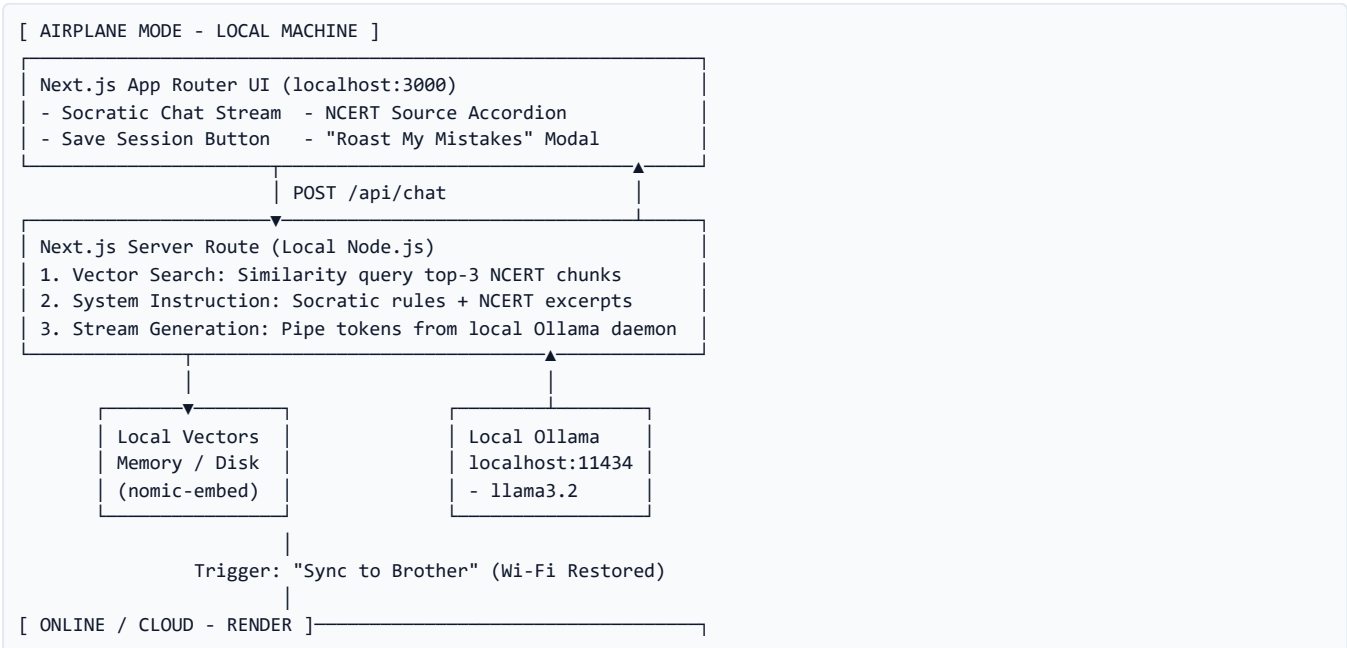
- The Distraction Loop:** Web-based studying invites social notifications and YouTube rabbit holes. Turning Wi-Fi off breaks standard web learning tools.
- The Spoon-Feeding Trap:** Standard conversational LLMs output complete answers directly, diminishing active recall and retention.
- Hallucination Risk:** Standard AI models invent scientific inaccuracies, which causes penalties in NEET's negative-marking scheme.

The Solution: A 100% offline desktop web app running via local open-source models. It utilizes local RAG strictly bound to NCERT textbooks, delivers Socratic hints, logs weak areas offline, and features a one-click cloud sync to a brother's review dashboard when Wi-Fi is re-enabled.

2. HACKATHON SCORING & STRATEGIC ALIGNMENT

Criterion	Weight	Implementation Strategy
Writing Quality	Highest	DEV post leads with an authentic sibling narrative, documented user reactions, architecture schematics, and terminal traces.
Open-Source AI Core	Mandatory	Local weights via Ollama (llama3.2 + nomic-embed-text). Justified by zero latency, zero internet requirement (focus), and privacy.
Theme Relevance	Mandatory	"Build for a Friend" — custom-engineered for the sister's daily study workflow and strict NCERT syllabus constraints.
Technical Execution	Standard	Full-stack TypeScript architecture, offline vector similarity search, streaming Web Streams, and local filesystem persistence.
Partner Technology	Bonus (\$200)	Sync receiver and brother review dashboard deployed live on Render .

3. HIGH-LEVEL ARCHITECTURE FLOW



```
Express + TypeScript Backend (Render Web Service)
- POST /api/sync -> Stores session & flagged weak points
- GET /dashboard -> Brother's visual progress review portal
```

4. DETAILED FUNCTIONAL SPECIFICATIONS

4.1 Offline RAG Pipeline (`/client/lib`)

- **Document Ingestion:** Local CLI script (`npm run ingest`) reading raw text/PDFs from `./data/ncert_docs/`.
- **Chunking:** `RecursiveCharacterTextSplitter` set to 500-token chunks with a 100-token overlap.
- **Embeddings:** Vectors generated via `ollamaEmbeddings` using local `nomic-embed-text`.
- **Vector Store:** Serialized index saved to `./data/vector_store/` via local `MemoryVectorStore`.

4.2 Socratic Inference & Prompts (`/client/app/api/chat`)

- Retrieve top-3 closest NCERT chunks ($k=3$).
- **Prompt Constraints:** Base hints strictly on the retrieved NCERT context. Never provide full answers or multi-step solutions. Provide one conceptual clue followed by one diagnostic question.
- Stream output via standard `ReadableStream` and `TextEncoder`.

4.3 Next.js Client User Interface (`/client/components`)

- **Airplane Mode Badge:** Visual confirmation: *"Offline Safe • Zero Network Calls"*.
- **Grounded Context:** Expandable accordion under messages showing exact NCERT chapter and source text.
- **Local Persistence:** "Save Session Offline" button serializing chat to `./data/study_log.json`.
- **Roast My Mistakes:** Analyzes session misconceptions in the voice of a teasing older sibling.
- **Cloud Sync Action:** Dispatches local JSON to Render when network connectivity returns.

4.4 Render Sync Server & Review Portal (`/server`)

- `POST /api/sync:` Accepts session payload and appends to `synced_sessions.json`.
- `GET /dashboard:` HTML report displaying session history and flagged weak topics for targeted mentoring.

5. DATA SCHEMA (`STUDY\_LOG.JSON`)

```
{
  "studentName": "Sanya",
  "timestamp": "2026-10-03T18:15:00.000Z",
  "subject": "Biology - Class 11",
  "chapter": "Cell: The Unit of Life",
  "metrics": {
    "totalInteractions": 12,
    "weakTopicsIdentified": [
      "Fluid Mosaic Model lipid bilayer symmetry",
      "Endomembrane system components"
    ]
  },
  "rawHistory": [
    { "role": "user", "content": "Why is the plasma membrane called quasi-fluid?" },
    { "role": "assistant", "content": "Think about the lipid bilayer. What component moves laterally within it to provide flexibility?", "source": "NCERT Ch 8, p. 131" }
  ]
}
```

6. HACKATHON SUBMISSION CHECKLIST

- ☐ Pull local models: `ollama pull llama3.2` and `ollama pull nomic-embed-text`
- ☐ Add sample NCERT chapter text into `./data/ncert_docs/` and run `npm run ingest`
- ☐ Turn laptop Wi-Fi off and verify chat inference runs with zero network connectivity
- ☐ Deploy `/server` to Render and confirm `GET /dashboard` loads online
- ☐ Have sister test the app for 15-20 minutes in Airplane Mode; photograph her using it and record her quote
- ☐ Publish DEV.to write-up before **Oct 5, 12:29 PM IST** with the open-source justification and Render link